

DARRELL VELENKOSINI NKOSI

Full-Stack Developer | React | Node.js | MongoDB | SQL
+27 063 465 0914 | velenkosininkosi02@gmail.com | [LinkedIn](#) | Bloemfontein / Open to Relocation

IBM Award - Best 1st-Year CIS, 2024 | BBD Award - Best 2nd-Year CIS, 2025 | Hult Prize 2nd Place, 2024 | Leadership Award, 2025

PROFESSIONAL SUMMARY

IBM & BBD Award-winning BCIS Honours student with hands-on experience building and analysing full-stack web applications using React, Node.js, and MongoDB. Built ClassBuddy - a live full-stack web application serving UFS students - applying end-to-end system design, API development, database architecture, and cloud deployment, with AI-assisted development using GitHub Copilot. Applied structured technical analysis to evaluate web application performance and user flows at Massmart (Makro), delivering actionable recommendations to senior stakeholders. Five officially certified UFS Demonstrator roles demonstrate deep technical knowledge and the ability to communicate complex concepts clearly. Seeking a junior Software Developer or IT Graduate role across South Africa.

TECHNICAL SKILLS

Languages: JavaScript (ES6+) - C# - Python - SQL - PHP
Frontend: React.js - Next.js - HTML5 - CSS3 - Responsive Design
Backend: Node.js - Express.js - REST APIs - .NET
Databases: MongoDB & MongoDB Atlas - SQL Server (T-SQL) - NoSQL
Deployment: Render (Backend) - Vercel (Frontend) - Git - GitHub
Tools: Power BI - Microsoft Visio - Draw.io - GitHub Copilot & AI Tools
Concepts: OOP - User Authentication & JWT - PDF Parsing - Calendar API Integration - Systems Analysis - UML - CRUD Operations

WORK EXPERIENCE

Full-Stack Developer - ClassBuddy

Personal Project - Live at [myclassbuddy.vercel.app](#) | 2026 - Present

Solo-built full-stack web application solving a real student problem - automating the manual, error-prone process of organising university timetables each semester. AI-assisted development using GitHub Copilot.

- Built a React.js frontend with component-based architecture, responsive design, and dynamic UI state management
- Developed a Node.js + Express.js backend with RESTful API endpoints, hosted and deployed on Render
- Implemented automated PDF parsing engine to extract and structure timetable data from raw university PDF documents
- Built real-time clash detection algorithm that identifies scheduling conflicts across selected modules
- Integrated Google Calendar, Outlook, and Apple Calendar export via calendar API
- Implemented full user authentication system with JWT-based session management and secure account flows
- Designed and managed MongoDB Atlas database for user data, timetable records, and feedback
- Deployed frontend on Vercel and backend on Render with environment variable management and production configuration

- Leveraged GitHub Copilot and AI-assisted development to accelerate feature delivery and code quality

React.js | Node.js | Express.js | MongoDB Atlas | JWT Auth | PDF Parsing | REST API | Calendar Integration | Vercel | Render | GitHub Copilot

A-STEP Tutor - CSIS1534 (C# Programming)

University of the Free State - Dept. CS & Informatics | 2026 - Present

- Facilitate weekly C# programming sessions teaching OOP, syntax, and debugging to first-year students
- Break down complex programming problems into structured steps, reinforcing own technical depth
- Track attendance and report progress to the supervising lecturer

C# Programming | OOP | Academic Support | Mentorship

UX Researcher & Digital Strategy Intern

Massmart (Makro) - Walmart Group | Jun 2025 - Jul 2025

Real-world industry internship - applied structured technical analysis to evaluate web application performance and user flows across Makro's e-commerce platform and mobile app.

- Applied structured analysis to evaluate web application performance and user flows across Makro's e-commerce platform
- Identified 10+ UX friction points across key user flows, informing prioritised platform improvement recommendations
- Delivered a professional technical report with data-backed recommendations and a 30-min stakeholder presentation
- Used HTML/CSS knowledge and digital systems understanding to analyse front-end and navigation issues

UX Research | Web App Analysis | Technical Reporting | Stakeholder Presentations | Journey Mapping

Demonstrator - Computer Literacy (CSIL1511, CSIL1561, CSIR1531 & CSIL1501)

University of the Free State | 2025 - Present

- Supported first-year students through Excel, Word, and PowerPoint practicals weekly
- Adapted explanations to diverse student backgrounds and skill levels

Microsoft Office Suite | MS Excel | MS Word | MS PowerPoint | Digital Literacy | Training & Instruction

Lab Assistant - Computer Science & Informatics

University of the Free State | 2025 Academic Year

Certified by the Head of Department - kept the CS lab running for hundreds of students across the academic year.

- Diagnosed and resolved hardware and software issues across lab stations in real time
- Supported students with Windows tools, coursework software, and lab equipment
- Coordinated with academic and technical staff on scheduling and lab operations

Technical Support | Hardware Troubleshooting | Windows Administration | Student Support | Lab Operations

Demonstrator - Introduction to Software Design (BCIS1623)

University of the Free State | 2025 Academic Year

Certified for BCIS1623 - teaching students to specify, model, and document business software systems.

- Guided students through UML diagrams (use case, class, activity, sequence) using Microsoft Visio
- Helped students translate business requirements into structured design documentation
- Managed mark-bearing attendance records and session logistics

UML Modelling | Systems Analysis | Business Requirements | Microsoft Visio | Documentation

Demonstrator - Systems Infrastructure & Integration (BCIS2624)

University of the Free State | 2025 Academic Year

Certified for BCIS2624 - covering enterprise systems infrastructure, networks, and integration patterns.

- Ran practicals on network diagram design using Draw.io and Microsoft Visio
- Explained how enterprise systems communicate and integrate in real business environments
- Managed mark-bearing attendance records as a formal administrative responsibility

Systems Infrastructure | Enterprise Integration | Network Diagrams | Draw.io | Microsoft Visio

Demonstrator - C# Programming (Introductory & OOP)

University of the Free State - Dept. CS & Informatics | 2024 - 2025 Academic Years | Certified by UFS

Certified across four C# modules - taking students from basic syntax through to OOP and Windows Forms GUI development.

- Taught C# fundamentals, OOP principles, and GUI development in hands-on practicals
- Ran live debugging sessions and promoted clean, secure coding practices
- Invigilated tests and supported lecturing staff across large student groups

C# / OOP | Debugging | Windows Forms / GUI | Visual Studio | Mentorship | Secure Coding

Corridor Prime (Ikhamanga) - Honorary Leadership Award

House Beyers Naude, UFS | 2025

- Awarded Honorary Award for Exceptional Leadership - signed by Residence Head & RC VP
- Managed corridor operations, resident welfare, conflict resolution, and access control
- Recognised for performance that went well beyond the standard expectations of the role

Leadership | Community Management | Conflict Resolution | Honorary Award

EDUCATION

Bachelor of Computer and Information Systems Honours (BCIS Honours)

2026 - Present

University of the Free State, Bloemfontein

Bachelor of Computer and Information Systems (BCIS)

2023 - 2025

University of the Free State, Bloemfontein

- IBM Award - Best First-Year CIS Student (April 2024)
- BBD Award - Best Second-Year CIS Student (April 2025)
- Hult Prize - 2nd Place, Innovation & Entrepreneurship Challenge (2024)

KEY PROJECT

ClassBuddy - Full-Stack Timetable Web Application

[Live App](#)

React.js + Node.js + MongoDB Atlas | Frontend: Vercel | Backend: Render

- Solo-built end-to-end - from system architecture and database design through to production deployment
- Automated PDF timetable parsing, real-time clash detection, group selection, calendar export, and user authentication
- AI-assisted development using GitHub Copilot, demonstrating productive use of modern developer tooling
- Built for UFS students - targeting 26,000 students across the university.

CERTIFICATIONS & SERVICE AWARDS

- Introduction to Business Analysis - LinkedIn Learning | IIBA Endorsed, 3.00 CDUs (Mar 2026)
- The Role of Business Analysis in Data Analytics - LinkedIn Learning (Mar 2026)
- Mastering AI Tools for C# Development - LinkedIn Learning (Sep 2025)
- Front-End Web Development with .NET - LinkedIn Learning (Sep 2025)
- Introduction to Web APIs - LinkedIn Learning (Sep 2025)
- Power BI: Dashboards for Beginners - LinkedIn Learning (Aug 2025)
- Introduction to Cybersecurity - Cisco Networking Academy (Jul 2025)
- UFS Certificates of Service (x7) - Dept. of CS & Informatics, UFS | BCIS1623, BCIS2624, CSIL1511, CSIL1561, CSIR1531, CSIL1501, CSIS1534/1644, CS Lab (certified by Head of Department)